Durva Sharma

durva.s170@gmail.com | Bengaluru, India | durva.xyz | github.com/sykeriin | linkedin.com/in/durva-s

I like to build the parts of AI systems that tend to break. I learn fastest by making projects. I'm deeply interested in AI research, while still exploring what else in tech genuinely excites me. Curious enough to stay up late reading a paper or rebuilding a system just to understand it, and looking to keep growing and eventually contribute meaningfully to AI research.

EDUCATION

B.Tech, Computer Science and Engineering | **Manipal Institute of Technology, Bengaluru**
Expected May 2029

WORK EXPERIENCE

Founding Software & AI Integration Engineer | **PetAlly**
Mar 2026 – Sep 2026 · The House of Vrindavan, Bengaluru

- Built PetAlly's recommendation system and integrated AI features across its website and mobile app, including an AI chatbot and pet mood journal.
- Developed pet-health tracking, engagement scoring, activity streaks and PawPoints rewards; contributed to app UI, landing-page development and AWS CI/CD.

OpenCV Preprocessing Lead | **HAWKEYE**
May 2026 – Present · Indian Air Force runway FOD project

- Led the computer-vision preprocessing pipeline for an Indian Air Force runway foreign-object-debris (FOD) detection system, integrating three Raspberry Pi camera feeds into a consistent stream of frames for downstream detection.
- Built a modular preprocessing and visualisation workflow (bilateral denoising, CLAHE, region-of-interest cropping, frame skipping) and structured the frontend for model integration and live feed inspection.

PROJECTS

tread — **Faster long-context prompts**

- It skips repeated prompt processing when documents are reused or rearranged, helping the model begin generating sooner.
- Implemented content-addressed GPU/CPU KV-cache tiers and exact RoPE re-rotation for reordered document chunks; on Qwen2.5-0.5B, 20% recomputation improved TTFT 1.4× at 4K and 1.6× at 8K context versus full prefill.
  Stack: Python, PyTorch, Hugging Face Transformers | **Code:** github.com/sykeriin/tread

SkillFuzz — **Fuzzing for agent tools**

- It checks what an agent's tools actually do, rather than trusting only the permissions they claim to need.
- Built a rootless sandbox with a network sinkhole and honeytokens that runs skills under many task and environment conditions, traces file, network and process effects, and diffs versions to flag undeclared capability growth; signed attestations and 44 tests.
  Stack: Python, Linux namespaces, strace | **Code:** github.com/sykeriin/SkillFuzz

DPDP Guard — **Privacy controls for AI apps**

- It gives developers a way to apply privacy rules consistently and gives users clearer control over how their data is handled.
- Built a consent SDK with per-purpose policy checks, pre-log PII masking and data-rights request detection (including Hinglish), backed by a tamper-evident audit ledger and dashboard.
  Stack: Python, FastAPI, PostgreSQL, Next.js | **Code:** github.com/sykeriin/dpdp-guard

TaintMem — **Trusted memory for agents**

- It preserves the source and trust level of information so an agent does not mistake an untrusted instruction for reliable memory.
- Implemented sentence-level trust labels, provenance propagation across LLM-style rewrites and policy-gated memory retrieval, designed to keep untrusted content from silently gaining authority.
  Stack: Python, SQLite | **Code:** github.com/sykeriin/TaintMem

CloudSense — **Cloud cost intelligence**

- It helps teams understand which services drive their cloud bill, spot unusual changes and make better cost decisions.
- Built and deployed a FinOps platform integrating AWS Cost Explorer, Isolation Forest anomaly detection and Prophet forecasting, with ML analytics, an AI assistant and what-if cost analysis.
  Stack: Python, FastAPI, React, AWS | **Code:** github.com/sykeriin/CloudSense

LEADERSHIP & ACHIEVEMENTS

- **Vice President, NeuraAI;** Working Committee, MAHE Innovation Centre.
- **1st place,** MAHE Mobility Hackathon 2026; **2nd place,** PowerNext (CPRI) and UI/UX Sprint, Tech Solstice 2026; **Top 10,** iQOO Hackathon 2026 and Point Blank CTF; **Top 32,** Zenith26 National Hackathon (1,000+ teams).

SKILLS & INTERESTS

Technical: Python, C++, TypeScript, JavaScript, SQL; PyTorch, Hugging Face Transformers, scikit-learn, OpenCV; FastAPI, React, Next.js, PostgreSQL, SQLite, AWS, Docker, Linux, Git.

Interests: Muay Thai, guitar, dance, fashion and visual design, poetry.